Umberto De Luca (he/him) **Computer Engineering**

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TECHNICAL SKILLS

- **Programming Languages:** Lua, Python, C#, Java, C, MATLAB
- **Frameworks/Libraries:** FastAPI, SQLAlchemy, Pydantic, TensorFlow, Scikit-learn, NLTK, JUnit, LÖVE2D
- **Development Tools:** Git/GitHub, Cursor/Codex, Unity, Visual Studio Code, Visual Studio, SolidWorks, Microsoft Office, Make

EDUCATION

University of British Columbia **2022 - Present**
Bachelor of Applied Science - Computer Engineering
CGPA: 80.0%
Related Courses:

- *Computer Programming I: 99%*
- *Computer Fluency: 91%*
- *Digital Logic Design: 90%*

Pontifícia Universidade Católica do Rio de Janeiro **2020 - 2022**
Mechanical Engineering
CGPA: 86%

TECHNICAL PROJECTS

Food Delivery Backend API, Course Project **February 2026 – April 2026**

- Designed and implemented a modular backend using MVC architecture, separating model, routing, controller, repository, and service layer logic using Python, VS Code and Git/GitHub
- Developed the order and delivery management system, including order creation, modification, and state transitions with robust validation using Pydantic
- Built a payment processing system combining a custom simulated transaction engine with the PayPal Sandbox API, handling transaction validation and edge cases to support realistic payment workflows and testing
- Wrote unit and integration tests using Pytest, applying equivalence partitioning and mock-based testing for reliability

Snake Game, Personal Project **June 2024 - July 2024**

- Developed a fully functional code and interface for a “Snake Game” with state machines for each game scene - start, play, and game over – using Lua, Visual Studio Code, and Love2d features
- Leveraged Lua's Knife library and its Timer.tween method to deliver an engaging gameplay experience by simulating a snake’s movement across the board grid
- Engineered a reliability layer around A* pathfinding for an autonomous Snake AI at the Start-State to significantly reduce self-trapping during autonomous gameplay
- Designed a modular, object-oriented architecture (Player, Board, GameObject, AI) using a custom Lua class system to separate game logic, rendering, and AI decision-making

OTHER WORK EXPERIENCE

Totaltag Soluções em Informática Ltda, Rio de Janeiro, Brazil **June 2026 – September 2026**
Technical Intern

- Mapped and documented the architecture and feature flows of a full-stack web application to support development and maintenance
- Developed and enhanced features for a dental clinic management platform using Python, FastAPI, Pydantic, SQLAlchemy with SQLite, React, and Pytest
- Leveraged AI-assisted development tools (Cursor, Codex, and large language models) to accelerate feature development, debugging, refactoring, and code analysis while validating generated code and maintaining software quality

University of British Columbia Food Services, Kelowna, Canada **September 2024 – April 2026**
Third Cook (Auxiliary)

- Collaborated with the cleaning staff to provide a safe and clean environment for customers
- Worked in a fast-paced environment, assisting the first and second cooks during their routines

Datasphere Dynamics, Vancouver, Canada **May 2025 – September 2025**
Intern

- Earned Microsoft M365 Fundamentals and Azure Fundamentals certifications as part of cloud training
- Supported customers in recovering compromised accounts and strengthening security posture, including endpoint hardening and account protection

Pontifícia Universidade Católica, Rio de Janeiro, Brazil **March 2021 - December 2021**
Linear Algebra & Calculus I Tutor

- Supported students who were struggling with Linear Algebra and Calculus I by solving and giving visual examples using MATLAB
- Collaborated with the course Professor to supervise students during examinations

ENGINEERING OR DESIGN STUDENT TEAMS

Reptiles-Baja, Pontifícia Universidade Católica do Rio de Janeiro **August 2021 - April 2022**
Suspension & Steering Team Member

- Collaborated with other members to build and design the suspension and steering system of the Baja Car using SolidWorks
- Performed practical and theoretical tests using MATLAB to understand and adjust the steering and suspension settings of the vehicle

VOLUNTEER EXPERIENCE

Pontifícia Universidade Católica, Rio de Janeiro, Brazil **March 2022 - July 2022**
Mathematics Assistant Professor at the "Seja Mais" Project

- Tutored a class of over 15 undergraduate mature students for university admissions exams
- Collaborated with the Mathematics Project Coordinator to provide an adequate curriculum

INTERESTS & ACTIVITIES

- **Completed the Introduction to Game Development and Introduction to AI free course version from Harvard edX - CS50**
- **Won the 2017 state and South American basketball U15 championship with "Clube de Regatas do Flamengo"**